

COURSE DESCRIPTION

Course Number	0418-223	Course Title	Systems Programming
Credit Hours	4 (3:3:4)	Course Coordinator	Dr. Hussain Almohri
Compulsory	Yes	Prerequisite	0418-221

Textbook

Title	Computer Systems: A Programmer's Perspective
Author	Randal E. Bryant and David R. O'Hallaron
Publisher	Pearson, 3 rd Edition, 2015

Title	An Introduction to Parallel Computing in C++
Author	Umut A. Acar and Arthur Chargueraud and Mike Rainey
Publisher	CMU, v1.2 2016-03 https://www.cs.cmu.edu/~15210/pasl.html

Major Topics Covered in the Course

	Topics	Hours
1	Linking : static linking, object files, symbol table and symbol resolution, symbol relocation, executable object files, static and dynamic linking, loading and linking libraries, position-independent code, library interpositioning	3
2	Exceptional Control Flow : exceptions, processes, system call error handling, signals, nonlocal jumps	6
3	Virtual Memory : physical and virtual addressing, address spaces, virtual memory as a tool, address translation, memory mapping, dynamic memory allocation, garbage collection	6
4	System-Level I/O : UNIX I/O, files, robust reading and writing, sharing files, I/O redirection, standard I/O	3
5	Networking Programming : the client-server model, physical networks, IP Internet, UNIX socket interface, web servers	6
6	Concurrent Programming : concurrency with processes, concurrency with I/O multiplexing, concurrency with threads, shared variables, semaphores, threads for parallelism, concurrency issues	6
7	Parallelism : fork-join parallelism, synchronization hardware, ABA problem, work efficiency, granularity control, determining threshold	6
8	Automatic Granularity Control : complexity functions, controlled statements, controlled parallel-for loop	3
	Exams	3
	Total	42

Laboratory Work:

	Topics	Week
1	Linking: object files, symbol resolution, relocation, static and dynamic linking	1
2	Exceptions, processes, and system call error handling	2
3	Process control and process manipulating tools	3
4	Sending and receiving signals, handling signals, non-local jumps, block and unblock signals	4
5	Virtual memory and memory mapping	5
6	Dynamic memory allocation and garbage collection: malloc(), free(), mark-and-sweep	6
7	System level I/O: Files, Directories, Meta-files	7
8	Network programming: IP addresses and domain names conversions, socket interface programming, and simple client/server	8
9	Tiny web server	9
10	Concurrent programming with: processes	10
11	Concurrent programming with: threads	11
12	Concurrent programming with: I/O multiplexing	12
13	Shared variables and Synchronizing process/threads: race conditions, semaphores	13
14	Experimenting with PASL	14

Grading

20%	Midterm 1
20%	Midterm 2
20%	Homework/quizzes/project
40%	Final

Course Outcomes

Students should be able to acquire the following skills.

1. Design and develop systems programs that use exceptional control flow and efficient I/O techniques.
2. Design and develop concurrent, parallel, and networked programs.
3. Analyze and understand basic problems in linking, virtual memory, and parallel programming.
4. Develop a systems program as part of a team project.
5. Generate technical reports for experimenting with systems programs and tools.

Relationship between Course Outcomes and Student Outcomes

Course Outcomes	Student Outcomes					
	1	2	3	4	5	6
1		H				
2		H				L
3	H					
4		H		L	H	
5	L		H			

Last updated Spring 2019